Gayathri Krishnamoorthy

Electrical Engineer| Computer Scientist

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Power Systems Engineer with 3+ years of experience as a researcher at Washington State University and Internship experience at Pacific Northwest National Laboratory (PNNL). Worked on multiple projects developing software tools to efficiently integrate renewable resources into power grid models. I am outgoing, creative, and detail-oriented with strong engineering basics and communication skills. I am effective both solo or on a team. In my spare time I practice yoga, meditation, and painting. I am also passionate about the brain sciences behind mindfulness and keeping up with recent developments in AI.

Education

Present – August 2018 | **Ph.D. in Electrical Engineering , Washington State University, Pullman, WA**

- Research : Transmission Distribution Co-simulation : Utility of Distribution Connected Renewable Resources in Providing Transmission System Support using Reinforcement Learning
- Major : Power Systems
- Minor : Computer Science
- Advisor : Dr. Anamika Dubey (WSU)
- Co-Advisors : Dr. Anjan Bose (WSU), Dr. Laurentiu Marinovici (PNNL)

May 2018 – Aug 2016 | **M.S. in Electrical Engineering, Washington State University, Pullman, WA**

- Research : An Iterative Co-simulation Framework for the Integrated Transmission and Distribution System Analysis
- Major : Power Systems
- Minor : Computer Engineering
- Advisor : Dr. Anamika Dubey (WSU)

May 2016 – Aug 2012 | **B.E. in Electronics and Communications Engineering, Anna University, Chennai, India**

- Title : Smart Home Design with Voice Recognition in a Single Controller (Arduino Uno) Environment
- Major : Embedded Systems
- Minor : Communications
- Advisor : Dr. Suresh T

Professional Experience

Present – August 2016 | **Graduate Research Assistant, Washington State University, Pullman, WA**

- Currently working on development of model-free power distribution system simulation for voltage control applications with probabilistic approaches and reinforcement learning techniques *(Python)*.
- Designed and developed a transmission and distribution (T&D) co-simulation platform to understand the impacts of increasing renewable penetrations on power system operations *(MATLAB, Python, and OpenDSS)*.
- Developed analytical models to demonstrate the convergence of T&D co simulation platform.
- Developed a hybrid T&D co-simulation platform that assists in secondary frequency regulation services to the power grid using distribution connected batteries *(PSAT, MATLAB, Python, and OpenDSS)*.
- Developed an optimization algorithm in a decentralized distribution market operation scenario to optimize the proposed market bids without violating system operating constraints *(MATLAB, OpenDSS)*.

Python | MATLAB | OpenDSS | PSAT | GridPV

Dec 2019
Aug 2019

Graduate Teaching Assistant, Washington State University, Pullman, WA
Course : Numerical Computing for Engineers
Tasks :
- Engaged a class of 80 undergraduate (multiple engineering discipline) students
- Helped the class in learning to use MATLAB, a professional computing software)
- Provided my best efforts in evaluating students' homeworks and exams

MATLAB

May 2019
Jan 2019

Graduate Teaching Assistant, Washington State University, Pullman, WA
Course : Professional Skills in Computing and Engineering
Tasks :
- Engaged a class of 108 undergraduate electrical and computer engineering students
- Taught the class to use Microsoft project professional software
- Detailed lectures on professional writing using LaTex.

Microsoft Project Professional | LaTex

Dec 2018
May 2018

Graduate Research Intern, Pacific Northwest National Laboratory, Seattle, WA
- Assisted in development of a transactive energy software agent, TESP, that optimizes the dynamics of distributed energy resources in a household and bids the excess energy in the energy market to provide monetary benefits to the costumers using electric vehicles, batteries, and PVs.
- Performed scalability testing for the open source co-simulation platform (HELICS) developed at PNNL in multiple programming language environments.
- Performed testing of the transactive energy simulation platform with HELICS that integrated multiple power system market components on a Linux environment
- Developed a C++ wrapper for MATPOWER software
- Developed a Docker Engine for the FNCS co-simulation software platform.

C++ | Python | MATLAB | MATPOWER | GridLAB-D | OpenDSS | Docker | EnergyPlus | FNCS | TESP | HELICS

Dec 2014

In-Plant Training, HCL Technologies, Chennai, India
- Developed fundamental principles and knowledge of recent advancements in Embedded Systems, Cloud Computing, Internet of Things, Android, and .Net software platforms.
- Attended robotics and embedded systems workshop on making of legged robots and successfully completed the project.
- Developed a prototype racing car that used Freescale controllers to automatically recognize the specially designed racetrack for the Freescale Cup intelligent car racing competition.

HTML | C | C++ | Python | Arduino | Raspberry Pi | EasyVR

Professional Activities

Present	Serving as a reviewer for IEEE Power and Energy Society Conferences, Institution of Engineering and Technology Journals, and IEEE Systems Journal
2019	Authored a Ph.D. Research Proposal to Pacific Northwest National Laboratory
2018	Paper Presentation- IEEE Power and Energy Society General Meeting
2017	Poster Presentation - Power Systems Engineering Research Center
2015	Attended Internet of Things with Arduino workshop for real-time applications at AltSense
2015	Trained and participated in FreescaleCup Intelligent Car Racing Competition

Programming Languages

C++	●●●●●
MATLAB	●●●●●
C	●●●●●
Python	●●●●○
HTML	●●●●○
Javascript	●●●○○
CSS	●●●○○

Simulation Tools

- Pytorch
- Keras
- Docker
- OpenDSS
- GridLAB-D
- Simulink
- MATPOWER

Projects

Transmission -Distribution Co-simulation 2017 - 2019

github.com/yvelaga/PSERC-Cosimulation PSERC T-60

Developed a framework for the integrated T&D system analysis to understand the impacts of distribution-connected renewable resources (DERs) on transmission system operations and vice-versa (project funded by PSERC). Also, using the co-simulation framework developed, explored the utility of DER's in transmission system support.

MATLAB | OpenDSS | Python

Real-time Control of Energy Storage Sytems for Frequency Regulation of Power Grid 2019

Energy Market

In this project, I am looking to explore the utility of DER's in transmission system support especially, frequency regulation. There are several aspects to this project : 1) Learning consumer willingness and flexibility in providing grid services using deep learning techniques and 2) Utilizing battery resources to provide combined active and reactive power support to mitigate the distribution grid violations while providing frequency regulation services.

MATLAB | OpenDSS | Python

Model-free Distribution System Development using Reinforcement Learning 2019

Machine Learning - Course Project

The goal of the project is to develop a model-free RL based voltage control using distributed energy resources. A policy gradient based reinforcement learning algorithm was implemented to control power distribution system voltage within operating limits using Torch.

Python | Torch

Distribution System Probabilistic Graph Model 2019

Data Science - Course Project

In this project, the conditional independence relationship of distribution system state variables in particular, voltages, is exploited under certain assumptions to build probabilistic graph model (PGMs). Here, PGM models the structured relationship among measurement variables(voltages) in one time-slice as a Gaussian graph model (GGM) and a chain graph representation is employed to capture the relationship between different time-slices. The broader goal of the project is to enable efficient management of grid's resources using the developed PGM.

Python

Website for Distribution lab at WSU 2019

DOPAL

Designed and maintain a professional website for my research group at WSU.

HTML | bootstrap | CSS | jquery

GMLC-TDC/ HELICS 2018

github.com/GMLC-TDC github.com/GMLC-TDC/MATPOWER-wrapper HELICS

HELICS is a coordinated effort by multiple National Labs to develop a platform that aids advanced power system operations. I worked on certain aspects of the development like scalability testing and wrapper development for multiple power system tools. One major takeaway during this internship was learning docker development.

C++ | Python | MATLAB | Matpower

TESP 2018

github.com/pnnl/tesp TESP

For this project, I was involved in developing the energy market structure (real-time and day-ahead) for HVAC (heating, ventilation and air conditioning) systems. The major tasks were to develop consensual bidding strategies and to test/validate on large scale power system networks.

Python | C++ | GridLAB-D | EnergyPlus

Real Time Cyber-Physical System Testbed Development 2017

Cyber Security - Course Project

Developed a real time cyber-physical system testbed communicating with each other through MatrikonOPC software to analyze the impacts of a real-time cyberattack on the power grid. Designed an Intrusion Detection System (IDS) in substations using SNORT software to provide additional security along with firewalls.

Matrikon OPC | MATLAB | Simulink | C++

Multi-core Processor Design 2016

github.com/adititihuse/Branch-Target-Buffer-Simulation

Development of an advanced multi-core processor design in C++ that handles functional units looping instructions for static and dynamic branch predictions

C++

Honors and Awards

2019 **Awardee** Graduate & Professional Student Association Scholarship Recipient for the Research Exposition at WSU
2018 **2nd Place** Best Student Paper Award at North American Power Symposium
2015 **Awardee** Completed course in CCNA Exploration : Network Fundamentals
2015 **2nd Place** Texas Instruments India Analog Maker Competition
2014 **Awardee** Level 1 Cambridge English Language Assessment in Business English
2011-2012 **Scholarship** 100 % fee concession in high school

Publications

2019 G. Krishnamoorthy and A. Dubey, "Transmission–Distribution Cosimulation : Analytical Methods for Iterative Coupling", in IEEE Systems Journal.
2019 J. C. Bedoya, C. Liu, G. Krishnamoorthy and A. Dubey, "Bilateral Electricity Market in a Distribution System Environment", in IEEE Transactions on Smart Grid.
2019 Sen, P.K., Velaga, Y.N., Chen, A., Krishnamoorthy, G. and Dubey, A., 2019," Advancements in Co-Simulation Techniques in Combined Transmission & Distribution Systems Analysis," The Journal of Engineering.
2019 G. Krishnamoorthy and A. Dubey, "Distributed PV Penetration Impact Analysis on Transmission System Voltages using Co-Simulation," accepted to appear in 2019 North American Power Symposium (NAPS), Wichita, 2019.
2019 G. Krishnamoorthy, A. Dubey and P. K. Sen, "Iteratively-Coupled Co-simulation Framework for Unbalanced Transmission-Distribution System," 2019 IEEE Milan PowerTech, Milan, Italy, 2019, pp. 1-6.
2018 Krishnamoorthy, Gayathri, and Anamika Dubey. "A framework to analyze interactions between transmission and distribution systems." In 2018 IEEE Power & Energy Society General Meeting (PESGM), pp. 1-5. IEEE, 2018.
2018 Y. N. Velaga, A. Chen, P. K. Sen, G. Krishnamoorthy and A. Dubey, "Transmission-Distribution Co-Simulation : Model Validation with Standalone Simulation," 2018 North American Power Symposium (NAPS), Fargo, ND, 2018, pp. 1-6.

References

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